Lean Canvas: DocuSense (hypotheses only; no market validation performed)

1. Problem: Invoices are checked by hand; arithmetic errors, duplicates and unusual amounts are easy to miss.

2. Customer segments (assumed): Small finance/accounts-payable teams; students and auditors learning document analysis. *Not validated with real customers.*

3. Unique value proposition: Free, local, explainable invoice screening that shows why a record needs review, with no paid APIs.

4. Solution: OCR + rule-based extraction, validation engine, fuzzy duplicate detection, optional anomaly detection, review dashboard.

5. Channels (proposed): Public GitHub repository, college/internship showcase, demo video.

6. Revenue streams: None for the academic prototype. Possible future: hosted version or support (unvalidated idea).

7. Cost structure: Developer time; open-source software; local hardware. No paid services.

8. Key metrics (measured & tracked):

- Field-level extraction on CORD v2 real receipts (100 docs): Subtotal precision 87.0% (recall 27.8%), Total precision 46.5% (recall 20.4%), Tax precision 44.4% (recall 8.9%).

- Validation coverage: 7 deterministic rules covering required fields, date sanity, due-date order, monetary plausibility, total consistency, tax ratio, and invoice number syntax.

- Anomaly model: Unsupervised Isolation Forest trained on 109 complete records (log_total, tax_ratio) with 5% contamination quantile.

- Test suite reliability: 51 automated unit/integration tests passing with 100% test success rate.

9. Unfair advantage: None claimed; differentiators are transparency, local privacy and zero licence cost.